

Inner Products and Eigenfunctions in Function Spaces

Joseph, Henry, Elizabeth

April 14, 2025

1 Introduction

This paper and the accompanying presentation review and extend concepts from linear algebra and differential equations. We first provide a general definition of the inner product and inner product spaces in $\mathbb{R}$. We then define eigenvalues and eigenvectors in $\mathbb{R}$, so that we can extend those ideas to function spaces as eigenfunctions. In the function space $C[a, b]$, we compute the eigenfunctions and corresponding eigenvalues associated with two common linear operators in the domain. Our final example uses the slightly more involved square integrable function space, $L^2[a, b]$. Taking the Laplacian as our linear operator, we derive the associated eigenfunction and eigenvalues and demonstrate the orthogonality of those eigenfunctions by extending the inner product into the infinite function space. We conclude with a brief reminder the reader of the relevance of this process to our in-class work with Spherical Harmonics.

2 Inner Products

2.1 What is an inner product in $\mathbb{R}$?

An inner product is an operation that combines two vectors in $\mathbb{R}^n$ to produce a single real number (a scalar). If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

then the inner product is:

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Key Points:

- The inner product is often called the dot product in $\mathbb{R}^n$.
- Vectors $\mathbf{u}$ and $\mathbf{v}$ are treated as $n \times 1$ matrices.
- The inner product is the matrix multiplication $\mathbf{u}^T \mathbf{v}$, resulting in a 1×1 matrix (a scalar).
- It measures how much two vectors "line up" with each other.
- Allows us to define:
 - Length of a vector
 - Angle between vectors
 - Orthogonality of vectors

2.2 What are inner product spaces in $\mathbb{R}$?

An **inner product** on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V , associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. $\langle c\mathbf{u}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle$
4. $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$, and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ iff $\mathbf{u} = \mathbf{0}$

A vector space with an inner product is called an **inner product space**.

2.3 Integral inner product

(For functions $f(x), g(x), h(x)$ over an interval $[a, b]$) The inner product is defined by:

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx$$

(a) Symmetry

$$\langle f, g \rangle = \int_a^b f(x)g(x) dx = \int_a^b g(x)f(x) dx = \langle g, f \rangle$$

(b) Additivity

$$\begin{aligned}\langle f + g, h \rangle &= \int_a^b (f(x) + g(x))h(x) dx \\ &= \int_a^b f(x)h(x) dx + \int_a^b g(x)h(x) dx \\ &= \langle f, h \rangle + \langle g, h \rangle\end{aligned}$$

(c) Scalar Multiplication

$$\begin{aligned}\langle cf, g \rangle &= \int_a^b (cf(x))g(x) dx \\ &= c \int_a^b f(x)g(x) dx \\ &= c\langle f, g \rangle\end{aligned}$$

(d) Positivity

$$\langle f, f \rangle = \int_a^b f(x)^2 dx \geq 0$$

- $f(x)^2 \geq 0$ for all real x
- $\langle f, f \rangle = 0$ if and only if $f(x) = 0$ almost everywhere

2.4 Sum inner product

(For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$)

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

(a) Symmetry

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

$$= v_1 u_1 + v_2 u_2 + \cdots + v_n u_n$$

$$= \langle \mathbf{v}, \mathbf{u} \rangle$$

(b) Additivity

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n (u_i + v_i) w_i$$

$$= (u_1 + v_1) w_1 + (u_2 + v_2) w_2 + \cdots + (u_n + v_n) w_n$$

$$= (u_1 w_1 + v_1 w_1) + (u_2 w_2 + v_2 w_2) + \cdots + (u_n w_n + v_n w_n)$$

$$= (u_1 w_1 + u_2 w_2 + \cdots + u_n w_n) + (v_1 w_1 + v_2 w_2 + \cdots + v_n w_n)$$

$$= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

(c) Scalar Multiplication

$$\langle c\mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n (cu_i) v_i$$

$$= (cu_1) v_1 + (cu_2) v_2 + \cdots + (cu_n) v_n$$

$$= c(u_1 v_1) + c(u_2 v_2) + \cdots + c(u_n v_n)$$

$$= c(u_1 v_1 + u_2 v_2 + \cdots + u_n v_n)$$

$$= c\langle \mathbf{u}, \mathbf{v} \rangle$$

(d) Positivity

$$\langle \mathbf{u}, \mathbf{u} \rangle = \sum_{i=1}^n u_i^2 = u_1^2 + u_2^2 + \cdots + u_n^2 \geq 0$$

Since each $u_i^2 \geq 0$, the whole sum is non-negative. If $\langle \mathbf{u}, \mathbf{u} \rangle = 0$, then $u_1 = u_2 = \cdots = u_n = 0$, thus $\mathbf{u} = \mathbf{0}$.

3 Eigenfunctions

3.1 What is an eigenvalue and an eigenvector in $\mathbb{R}$?

An eigenvector of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar λ .

- The scalar λ is called an **eigenvalue** of A .
- The vector $\mathbf{x}$ is called an **eigenvector** corresponding to λ .

3.2 Eigenfunctions and Function Spaces

Recall from linear algebra the concepts of eigenvalues and eigenvectors. As mentioned earlier, the standard equation for eigenvectors is $A\mathbf{x} = \lambda\mathbf{x}$, where A is a matrix, $\mathbf{x}$ is an eigenvector, and λ is the corresponding eigenvalue. When discussing eigenfunctions, a similar form applies: $L(f) = \lambda f$, where L is a linear operator acting on a function f . In this context, we use linear operators instead of matrices to explore how functions behave under transformation. The specific cases discussed in more detail are in the space of $L^2[a, b]$, the space of squared integral functions. These are nonzero functions that when acted upon by a linear function result in a scalar multiple of itself, known as the eigenvalue. Using eigenfunctions allows us to look at problems related to infinite-dimensional functions that we were unable to look at with eigenvectors.

3.3 Function Space ($\mathbb{R}$)

Starting off looking at a basic example in $\mathbb{R}$. Let us say that we have the equation $L(f) = f'$. In order to solve this problem we use the equation discussed above such that $L(f) = \lambda f$. Plugging in the linear operator equation, we get

$$f' = \lambda f$$

Next, we divide both sides by f and take the integral of both sides.

$$\begin{aligned}\frac{f'}{f} &= \lambda \\ \int \frac{f'}{f} dx &= \int \lambda dx \\ \ln |f(x)| &= \lambda x + C \\ f(x) &= Ce^{\lambda x}\end{aligned}$$

Thus, the general solution for the eigenfunction is $f(x) = Ce^{\lambda x}$

3.4 Eigenvalue Problem 1 in $C[a, b]$

Now looking at a more complex example in $C[a, b]$. Using an example of an integral operator with a constant kernel, let us examine the problem of $T(f)(t) = \int_0^1 f(s) ds$ with $C \in [0, 1]$. Using the eigenvector equation we get $Tf(t) = \lambda f(t)$. Now plugging in our equation results in

$$\int_0^1 f(s) ds = \lambda f(t)$$

Notice that the left-hand side is just a constant, while the right-hand side depends on x . This means that the only way for the equation to be valid for all t is for $f(t)$ to be a constant function so let's assume that $f(x) = c$ where c is some constant. Then substituting this back in we get:

$$\begin{aligned}\int_0^1 ds &= \lambda c \\ c &= \lambda c \\ \lambda &= 1\end{aligned}$$

Thus the eigenvalue is $\lambda = 1$ and the eigenfunction is any constant function for some constant c

3.5 Eigenvalue Problem 2 in $C[a, b]$

Moving on to look at a problem in the function space of $C[a, b]$. We are going to focus on a specific known as LTI systems. An LTI system, or Linear Invariant systems. This system is used mostly in electrical physics in order to model and analyze how systems respond to inputs. The specific problem that we'll be looking at is $H(x(t)) = \int_{-\infty}^{\infty} h(\tau) * (t-\tau) d\tau$ considering the input $x(t) = e^{st}$ with $s \in \mathbb{C}$. Plugging $x(t) = e^{st}$ into our original equation we get:

$$\begin{aligned} H(e^{st}) &= \int_{-\infty}^{\infty} h(\tau) * e^{s(t-\tau)} d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) * e^{st} * e^{-s\tau} d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau) * e^{-s\tau} d\tau \\ H(e^{st}) &= \lambda s e^{-s\tau} \end{aligned}$$

Thus our eigenvector is equal to e^{st} and our eigenvalue is equal to $\int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$. This is also the Laplace transform, which converts a function of time into a function of complex variables.

4 Inner Products and Eigenfunctions in Function Spaces

4.1 Function Space $L^2[a, b]$

$L^2[a, b]$ is the space of all integrable squared functions. Functions in the space satisfy $\int_{-\infty}^{\infty} (|f(x)|)^2 dx < \infty$.

4.2 Orthogonality of Eigenfunctions using Inner Product

Recall that eigenfunctions f satisfy the equation $L(f) = \lambda f$ where L is a linear operator, f is an eigenfunction, and λ is an eigenvalue. Consider an eigenvalue problem with the Laplace Operator on Dirichlet boundary conditions in the infinite function space $L^2[0, \pi]$. Dirichlet conditions specify $y(0) = y(\pi) = 0$ and the Laplace Operator is defined as

$$L(f) = -\frac{d^2 y}{dx^2}$$

We solve for positive eigenvalues and the associated eigenfunctions. Then $L(y) = \lambda y = -y''$ and

$$y'' + \lambda y = 0$$

with $y(0) = y(\pi) = 0$. Because our equation is a linear, second order homogeneous differential equation, we have two results. First, the general characteristic equation is $ar^2 + br + c = 0$, and thus, the characteristic equation for our Laplace Operator equation is:

$$\begin{aligned} r^2 + \lambda &= 0 \\ \implies (r + \sqrt{\lambda}i)(r - \sqrt{\lambda}i) &= 0 \\ \implies r &= \pm \sqrt{\lambda}i \end{aligned}$$

Second, the general solution is of the form $y(t) = c_1 e^{\phi t} \cos(\mu t) + c_2 e^{\phi t} \sin(\mu t)$ for $r = \phi \pm \mu i$ [1]. Then the general solution to our equation is $y(t) = c_1 e^{\phi \cdot 0} \cos(\sqrt{\lambda} t) + c_2 e^{\phi \cdot 0} \sin(\sqrt{\lambda} t)$. Simplified:

$$y(t) = c_1 \cos(\sqrt{\lambda} t) + c_2 \sin(\sqrt{\lambda} t).$$

We now plug in the Dirichlet boundary conditions $y(0) = y(\pi) = 0$:

$$\begin{aligned} y(0) &= c_1 \cos(\sqrt{\lambda} \cdot 0) + c_2 \sin(\sqrt{\lambda} \cdot 0) = 0 \\ &= c_1 = 0 \end{aligned}$$

$$\begin{aligned} y(\pi) &= c_1 \cos(\sqrt{\lambda} \cdot \pi) + c_2 \sin(\sqrt{\lambda} \cdot \pi) \\ &= c_2 \sin(\sqrt{\lambda} \cdot \pi) = 0 \end{aligned}$$

$$y(t) = c_2 \sin(\sqrt{\lambda} \cdot t)$$

Since we specified positive eigenvalues, $c_2 \neq 0$ and $\sin(\sqrt{\lambda} \cdot \pi) = 0$. These relationships force the following result for λ with $n \in \mathbb{N}$ [4]:

$$\sqrt{\lambda_n} = n$$

$$\lambda_n = n^2$$

This general eigenvalue form yields the following eigenfunction solution:

$$y_n(t) = \sin(n \cdot t)$$

For Hermitian operators like Laplace, the eigenfunctions are orthogonal if they correspond to different eigenvalues [2]. [Note that Hermitian operators are linear operators on a Hilbert space, which is essentially an extension of a two-dimensional Euclidean space into a complete, three-dimensional space.] We will prove the orthogonality of our example eigenfunctions by extending the concept of inner product to function spaces. Note that all eigenfunctions y_n are members of the function space L^2 , since the integral of the square of $\sin(x)$ is finite. Then we can use the L^2 inner product, defined for $f, g \in L^2$ over region Ω as:

$$\langle f, g \rangle = \int_{\Omega} f(x)g(x) dx$$

Choose $n_1, n_2 \in \mathbb{N}$ and $n_1 \neq n_2$. Then we have the following eigenvalue and eigenfunction pairs:

$$\begin{aligned} \lambda &= n_1^2 \\ y_{n_1} &= \sin(n_1 t) \end{aligned}$$

$$\begin{aligned} \lambda &= n_2^2 \\ y_{n_2} &= \sin(n_2 t) \end{aligned}$$

Recall the trigonometric identity $\sin(A)\sin(B) = \frac{1}{2}(\cos(A - B) - \cos(A + B))$. Use the identity to take the inner product of the above general eigenfunctions of the Laplace Operator:

$$\begin{aligned}
\langle \sin(n_1 t), \sin(n_2 t) \rangle &= \int_0^\pi \sin(n_1 \cdot t) \cdot \sin(n_2 \cdot t) dt \\
&= \int_0^\pi \frac{1}{2} (\cos(n_1 t - n_2 t) - \cos(n_1 t + n_2 t)) dt \\
&= \frac{1}{2} \int_0^\pi \cos(n_1 t - n_2 t) dt - \frac{1}{2} \int_0^\pi \cos(n_1 t + n_2 t) dt \\
&= \frac{1}{2(n_1 - n_2)} \sin(t(n_1 - n_2)) \Big|_0^\pi - \frac{1}{2(n_1 + n_2)} \sin(t(n_1 + n_2)) \Big|_0^\pi \\
&= 0
\end{aligned}$$

Justification for the transition from the fourth line to the zero solution:

Consider that, since n_1 and n_2 are $\in \mathbb{N}$, then $n_1 - n_2$ and $n_1 + n_2$ are $\in \mathbb{Z}$. Then for upper boundary π , the argument of both $\sin$ terms is a positive integer multiple of π , and $\sin$ maps the argument to 0, zeroing out the entire term. For the lower boundary 0, the argument of $\sin$ is 0 and all terms with $\sin$ therefore equal 0. Then all terms are 0 and the inner product is equal to 0. Therefore, by definition, the eigenfunctions of the Laplace Operator are orthogonal! It is possible to further prove that the eigenfunctions form an orthogonal basis for $L^2[0, \pi]$, but this additional step is beyond the scope of this report.

4.3 Proving orthogonality of Spherical Harmonics

Spherical harmonics are themselves the eigenfunctions of the angular component of the three-dimensional Laplace Operator [3]. Recall that we proved in-class that two spherical harmonics of different degrees are orthogonal over the sphere, i.e. for $n \neq m$:

$$\langle Y_n(\zeta), Y_m(\zeta) \rangle = \int_{S^2} Y_n(\zeta) \cdot Y_m(\zeta) dS = 0$$

The following is a rough conceptual outline of the proof (please refer back to class notes for a rigorous proof). Substitute the harmonic, homogeneous polynomial H_n and H_m for Y_n and Y_m , respectively. Apply the divergence theorem to $H_n \nabla H_m$ and $H_m \nabla H_n$ and subtract the results. A series of convoluted algebraic steps involving the product rule for divergence, Euler's Theorem, and properties of harmonic polynomials yield the above relationship, and Y_n and Y_n are orthogonal.

References

- [1] Paul Dawkins. “Second Order DE’s”. In: *Paul’s Online Notes* (2025).
- [2] Richard Fitzpatrick. “Eigenfunctions of Operators are Orthogonal”. In: *Chem LibreTexts* (2025).
- [3] Maxwell Thum Matt DeCross and Jimin Khim. “Spherical Harmonics”. In: *Brilliant.org* (2025).
- [4] William F. Trench. *Elementary Differential Equations with Boundary Value Problems*. Faculty Authored and Edited Books CDs, 2013.